Comparison of Proposed Lockdown Models for COVID-19 Second Wave in India

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Introduction

- India reported 3,78,075 new cases of COVID-19 (on 5 May 2021,) accounting for 47% of new cases reported globally and 276 daily cases per million population.
- Presently, India is going through the second wave with so many morbidity, mortality with scarcity of oxygen supply, hospital beds, and supply of essential medicines throughout the country.
- India were able to combat the first wave of COVID-19 in 2020 successfully, where complete lockdown followed by so many restrictions were implemented very strictly.
- In this year, complacency has been observed in maintaining COVID-19 appropriate behaviour by the people.
- Already so many initiatives have been undertaken like COVID-19 vaccination, installation of new oxygen plants, conversion of nitrogen generator to oxygen generator etc.
- In spite of the different initiatives daily number of COVID-19 cases stands around more than 3.5 lakh cases per day. Some authorities have already raised their voice in favour of lockdown despite adverse consequences in terms of economic outcome.
- In this background, the present study was conducted to check out the different options for lockdown to tackle the transmission of the disease.

Objective

- To forecast the estimated number of COVID-19 cases per day in proposed different lockdown models in our country over a period for next 50 days i.e. from 10th May – 30th June, 2021.

Materials & Methods

- **Data source:**
- The study was conducted with the help of secondary data set obtained from Coronavirus Source Data, which is available at https://covid.ourworldindata.org/data/ecdc/total_cases.csv last accessed on 4th May, 2021.
- **Models**
- Auto regressive models (AR), Moving Average models (MA), Auto regressive Moving Average models (ARMA), and Autoregressive Integrated Moving Average model (ARIMA) was used for forecasting.
- **Following assumptions are taken:**
- Number of persons in home quarantine, Asymptotic patients with positive cases, government quarantine places, self-isolation, along with lag of test procedure, reduction in government transport, timing of market places, were taken into assumptions to fit the ARIMA model.

Results

- In absence of blanket lockdown in India, daily reported cases of COVID-19 will go up initially and will then gradually come down to 2,869,64 as on 30th June, 2021.
- If complete lockdown is implemented for 25 days from 6th May to 31st May, 2021 in 6 most affected states, then gradually the daily cases will be reduced from 10th May, 2021 onwards.
- Again, if blanket lockdown is implemented for the entire country for the same duration i.e. 6th May to 31st May, 2021, then drastically the daily reported cases will be decreased from 2,457,12 (10th May, 2021) to 80,469 (30th June, 2021) (Table 1).

Table 1: Predicted daily confirmed cases of COVID-19 in India from 10th May to 30th June, 2021 through forecasting in various lockdown models

Date (DD-MM-YEAR)	There are no Lockdown	Complete 25 days lockdown in 6 most affected states *	Blanket lockdown in India **
	Daily affected confirmed cases Number (95% Confidence Interval)	Daily affected confirmed cases Number (95% Confidence Interval)	Daily affected confirmed cases Number (95% Confidence Interval)
10-05-2021	3,769,84 (3,750,95 - 3,788,73)	3,182,43 (3,169,84 - 3,195,02)	2,457,12 (2,440,28 - 2,473,96)
15-05-2021	4,169,21 (4,150,32 - 4,188,10)	3,094,85 (3,082,26 - 3,107,44)	2,158,96 (2,142,12 - 2,175,80)
20-05-2021	3,981,72 (3,962,83 - 4,000,61)	2,831,89 (2,819,30 - 2,844,48)	1,905,63 (1,888,79 - 1,922,47)
31-05-2021	4,075,62 (4,056,73 - 4,094,51)	2,632,36 (2,619,77 - 2,644,95)	1,642,93 (1,626,09 - 1,659,77)
10-06-2021	3,617,64 (3,598,75 - 3,636,53)	2,027,08 (2,014,49 - 2,039,67)	1,348,69 (1,331,85 - 1,365,53)
20-06-2021	3,016,72 (2,997,83 - 3,035,61)	1,413,08 (1,400,49 - 1,425,67)	1,123,69 (1,106,85 - 1,140,53)
30-06-2021	2,869,64 (2,850,75 - 2,888,53)	1,070,90 (1,058,31 - 1,083,49	80,469 (78,785 - 82,153)

* Complete or blanket lockdown for 25 days (6th May to 31st May, 2021) in those states from where maximum number of positive cases is reported like West Bengal, Maharashtra, Delhi, Haryana, Rajasthan, and Telengana.

** Complete or Blanket lockdown for 25 days (6th May to 31st May, 2021) over the country.

Conclusion

- Maximum gain in terms of reduction of new cases will be achieved in complete lockdown model throughout the country.
- Particular state wise lockdown model will also decrease number of new cases considerably.

Recommendation

Although the blanket lockdown model seems to be best in our country, but this model is very aggressive in terms of economic turnover. Particular state wise lock down model will also be beneficial to certain extent.

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